
```
clc;
clear;
close all;

% Parameters
Am = 1;
Ac = 1;
fm = 5;
fc = 50;

beta = 5;          % FM modulation index
kp = 5;           % PM sensitivity

fs = 1000;
t = 0:1/fs:1;

% Message Signal
m = Am*cos(2*pi*fm*t);

% Carrier Signal
c = Ac*cos(2*pi*fc*t);

% FM Signal
fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

% PM Signal
pm_sig = Ac*cos(2*pi*fc*t + kp*m);

% Frequency Deviation
Df = beta*fm;

% PM Modulation Index
PM_index = kp*Am;

% Display Results
fprintf('FM Frequency Deviation = %.2f Hz\n', Df);
fprintf('FM Modulation Index = %.2f\n', beta);
fprintf('PM Modulation Index = %.2f\n', PM_index);

% Plot Results
figure('Name', 'FM and PM Modulation', 'NumberTitle', 'off');

subplot(4,1,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,2);
plot(t,c);
title('Carrier Signal');
```

```
xlabel('Time (s)');  
ylabel('Amplitude');  
grid on;
```

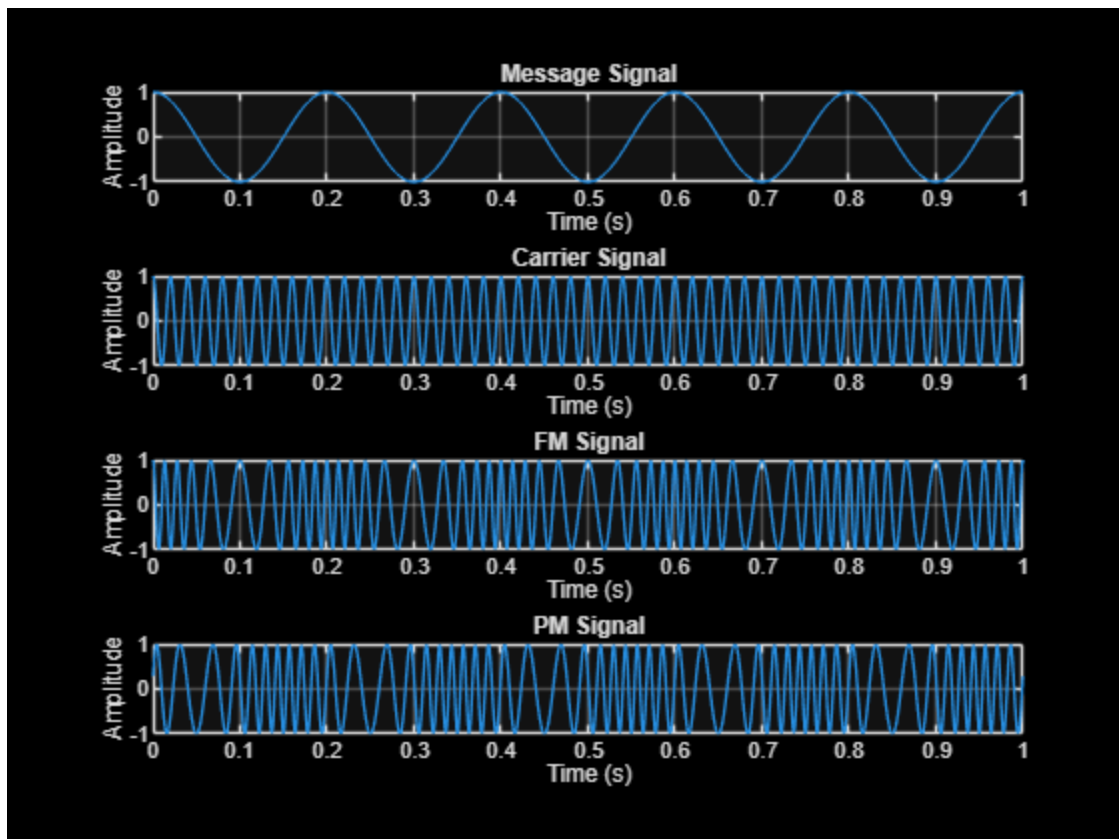
```
subplot(4,1,3);  
plot(t, fm_sig);  
title('FM Signal');  
xlabel('Time (s)');  
ylabel('Amplitude');  
grid on;
```

```
subplot(4,1,4);  
plot(t, pm_sig);  
title('PM Signal');  
xlabel('Time (s)');  
ylabel('Amplitude');  
grid on;
```

FM Frequency Deviation = 25.00 Hz

FM Modulation Index = 5.00

PM Modulation Index = 5.00



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